

Avengers: Endgame — Social Media Analytics Framework

A structured analysis of 15,000 tweets collected during a concentrated event window on 23 April 2019.

15,000

RECORDS

90%

RETWEETS

~81 min

TIME WINDOW

10,105

UNIQUE USERS

Prepared for the Social Media Analytics Framework prototype · SIH problem statement

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Key Takeaways

- The conversation is overwhelmingly retweet-driven (90%), making it ideal for content-propagation analysis.
- All activity occurred within a single ~81-minute event window — treat findings as a snapshot, not a trend over time.
- Participation is broad (10,105 users) but shallow (max 49 records/user), with replies too sparse for full network analysis.

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1. Purpose of This Dataset

This Twitter dataset will be used as the main dataset for the Social Media Analytics Framework prototype. The dataset contains Twitter activity related to Avengers: Endgame during a short period on 23 April 2019.

Our prototype will use this data to demonstrate four main capabilities:

- Timeline Analysis — understand when and how actively people were talking.
- Sentiment Analysis — understand whether the conversation was positive, neutral, or negative.
- Trend Analysis — identify popular hashtags, keywords, and topics.
- Content Propagation Analysis — understand which content was being amplified through retweets.

Scope note

The dataset is not large enough or long enough to represent the entire Avengers: Endgame conversation. It should instead be treated as a snapshot of a highly active online conversation during a specific event window.

2. Dataset Overview

The dataset contains a large number of users but relatively few tweets per user. The most active user has 49 records. This means the dataset represents participation from a large number of users rather than being dominated by a small group of users.

3. Original Dataset Fields

The original dataset contained 17 columns.

Important fields

A focus was placed on the fields required for the analytical prototype (see later sections for usage).

Fields not required for the prototype

The following fields were removed from the working dataset:

Removed field	Reason
index	CSV row number.
favorited	Not useful for our current analysis.
truncated	Not required for the prototype.
replyToSID	Technical reply ID not required for the current analysis.
replyToUID	Technical user ID not required for the current

	analysis.
statusSource	Application/device used to post.
retweeted	Constant in this dataset.
longitude	Only four records contain values.
latitude	Only four records contain values.

The original CSV is kept unchanged, so these fields can be recovered later if needed.

4. Important Findings From Exploration

4.1 The dataset is heavily dominated by retweets

Out of 15,000 records:

- 13,501 are retweets.
- 1,499 are original tweets.

Key finding

90% of the dataset consists of retweets. This is one of the most important characteristics of the dataset — the conversation was strongly focused on amplifying existing content rather than continuously creating new content, making it particularly useful for studying content propagation.

4.2 The conversation covers only a short period

The earliest record is 2019-04-23 09:22:01. The latest record is 2019-04-23 10:43:30. Therefore, all 15,000 records occur within approximately 81 minutes.

This means we should not claim that the dataset shows the evolution of the Avengers: Endgame conversation over days or weeks. Instead, our analysis will focus on:

- Conversation dynamics within a concentrated event window.

4.3 Twitter activity is consistently high

Tweets were grouped into five-minute intervals. The number of retweets remained much higher than the number of original tweets throughout the entire period. There is no single extreme activity spike in this sample; instead, the dataset shows consistently high activity with strong retweet amplification.

5. Repeated Content and Retweets

The dataset contains 4,656 unique tweet texts among 15,000 records. Some texts occur many times because the same content was retweeted by many users.

For example, the most frequently occurring text appears 1,456 times: “RT @HelloBoon: Man these #AvengersEndgame ads are everywhere...”

Other highly repeated content includes posts from accounts such as:

- @Avengers
- @Marvel
- @MarvelStudios
- @MCU_Direct
- @TicketNew

This gives us an opportunity to identify which narratives and pieces of content were being amplified most frequently in our sample.

Important distinction

Do not double-count

retweetCount and the number of times a tweet appears in our dataset are different measurements. retweetCount represents the recorded popularity of the underlying tweet, while repeated appearances of the same text represent how often that content appears in our collected sample. Therefore, we should not simply add all retweetCount values together and describe the result as total retweets.

6. User Distribution

There are 10,105 unique users across the 15,000 records. The most active users have only a few dozen records; the most active user has 49 records. This indicates that participation is spread across many users.

For the prototype, we can therefore investigate:

- Most active users.
- Users associated with highly propagated content.
- Accounts whose content receives high engagement.
- Relationships between users where interaction data is available.

7. Reply Data

Only 397 records contain a reply target. This represents approximately 2.65% of the dataset. Therefore, a traditional reply-based social network will be relatively sparse.

Design decision

Because of this, detailed network analysis will be treated as an optional extension for the prototype. Instead, retweet-based content propagation will be a stronger initial way to demonstrate how information spreads.

8. Data Cleaning and Feature Engineering

A cleaned dataset was created from the original data. The original CSV was preserved separately. The cleaned dataset contains 14 columns.

9. Text Cleaning

The original text was preserved, and a `text_clean` column was created by removing URLs, normalizing whitespace, and stripping line-break artifacts. Hashtags, mentions, emojis, and repeated punctuation were intentionally kept as useful signals for sentiment and trend analysis; further preprocessing happens inside the individual NLP components.

10. Timeline Features

The timestamp was converted from a string into a proper datetime value. Additional time features were created:

date

The calendar date of the tweet.

hour

The hour in which the tweet was posted.

minute

The minute in which the tweet was posted.

time_bin

The timestamp rounded down to a five-minute interval. For example, 2019-04-23 09:22:01 becomes 2019-04-23 09:20:00.

This makes it easier to calculate:

- Tweets per five minutes.
- Retweets per five minutes.
- Original tweets per five minutes.
- Sentiment per five minutes.
- Trending topics over time.

11. Tweet Type Feature

The original boolean field `isRetweet` was converted into a more readable field `tweet_type` with two possible values:

- original
- retweet

This makes analysis and visualization easier.

12. Data Quality Validation

Post-cleaning checks confirmed: all 15,000 records were retained (no accidental loss), 0 duplicate tweet IDs exist, and key fields (tweet ID, text, timestamp, user, retweet status, engagement) contain no missing values. The only notable gap is reply_to_user (14,603 missing), expected given only 397 replies. The retweet/original split (13,501 / 1,499) was preserved.

13. How This Dataset Will Be Used in the Prototype

The cleaned Twitter dataset will serve as the common input for several analytical components.

A. Timeline Analysis

Input: created, time_bin, is_retweet

- Tweet volume over time.
- Original vs retweet activity.
- Periods of higher/lower activity.

B. Sentiment Analysis

Input: text_clean

- Sentiment label.
- Sentiment score.
- Positive / neutral / negative distribution.
- Sentiment over time.
- Potential comparison between original tweets and retweets.

For the first prototype, we will use a simple baseline sentiment model before attempting more advanced models.

C. Trend Analysis

Input: text, and later extracted hashtags, keywords, and topics.

- Most discussed hashtags.
- Most common keywords.
- Popular topics.
- Topic frequency over time.
- Potentially rising topics.

D. Content Propagation Analysis

Input: text, is_retweet, retweet_count, user

- Most frequently repeated content.
- Highly propagated narratives.
- Popular source accounts.
- Engagement associated with propagated content.

Priority component

This component will be particularly important because approximately 90% of the dataset consists of retweets.

14. Limitations

Short time window

The dataset covers only around 81 minutes, so it cannot show long-term changes in sentiment or trends.

Historical dataset

This is not live Twitter data. The final system proposed for the SIH problem would require a continuous data ingestion pipeline, but our prototype uses a historical CSV to demonstrate the analytical components.

Limited reply information

Only 397 records contain reply information, so detailed reply-network analysis may be limited.

No demographic information

The dataset does not provide reliable age, gender, profession, or geographic information for most users. Therefore, the demographic profiling component of the full SIH solution cannot be properly demonstrated using this dataset alone.

Retweet-heavy data

Because 90% of the records are retweets, raw tweet counts should not automatically be interpreted as the number of independent opinions. Repeated content represents amplification and should be handled separately from original content.